

Firewall Project

안응렬

권택

박소정

윤승원

황준서

01 방화벽 구성도

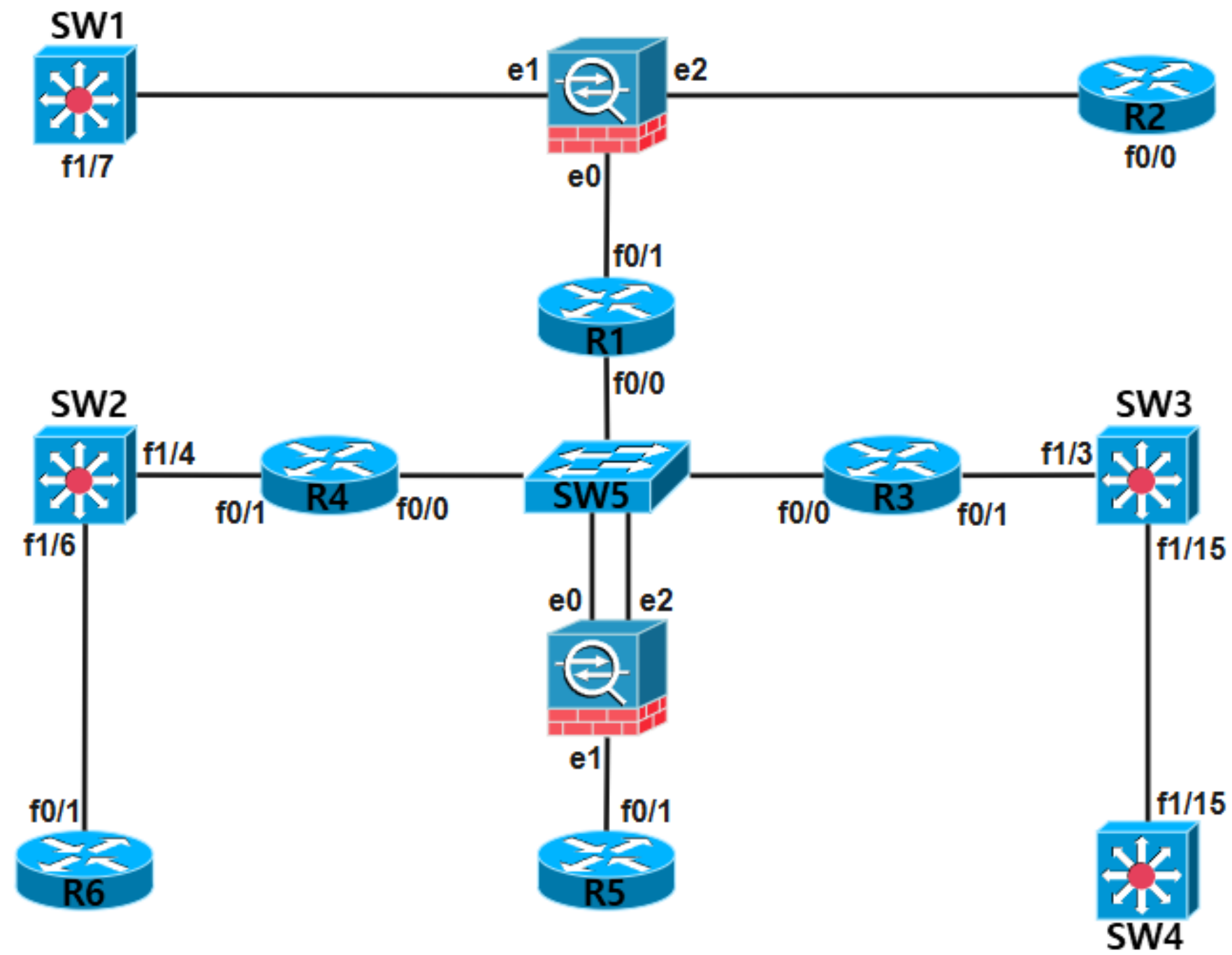
- 1-1. 물리적 구성도
- 1-2. 기능적 구성도
- 1-3. IGP 구성도

02 장비 설정

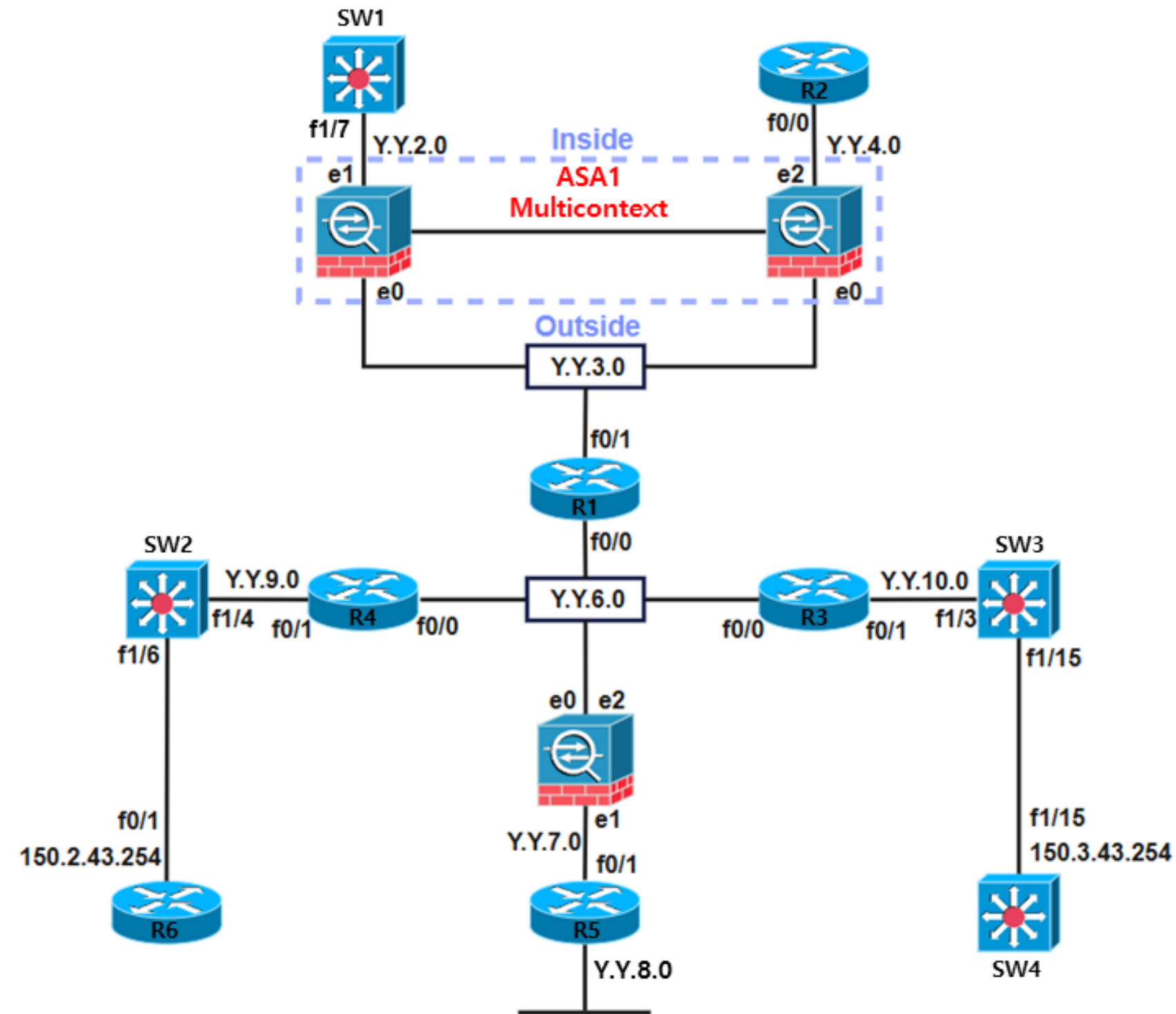
- 2-1. Router 설정
- 2-2. Switch 설정
- 2-3. Firewall 설정
- 2-4. Firewall (NAT) 설정
- 2-5. Routing Table 확인

01 방화벽 구성도

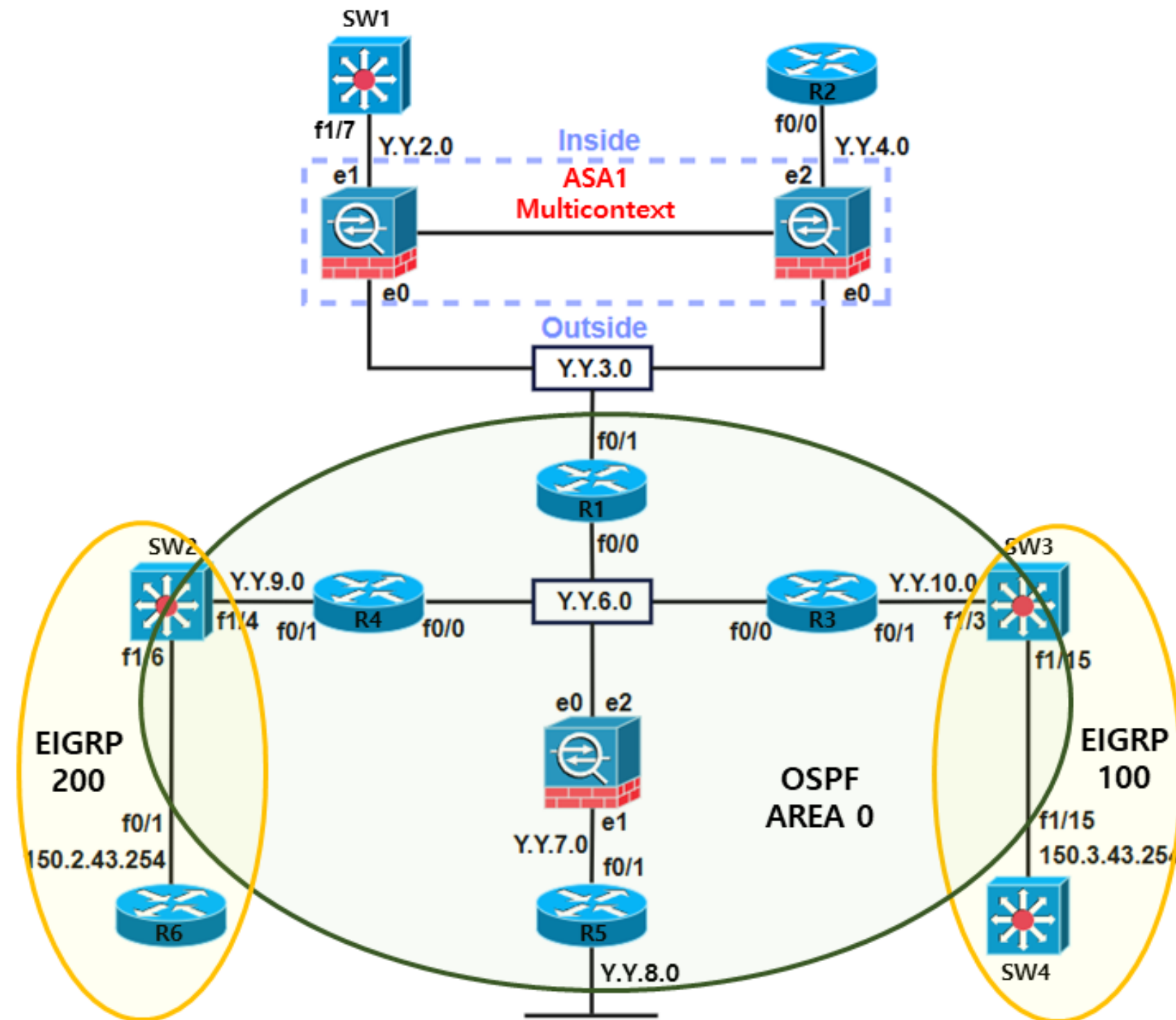
물리적 기능도



논리적 기능도



IGP 기능도



02 장비 설정

Router 설정

R1

```
int lo0  
ip add 192.168.1.1 255.255.255.255
```

```
int lo2  
ip add 43.43.51.1 255.255.255.255
```

```
int f0/0  
no sh  
ip add 43.43.6.1 255.255.255.0
```

```
int f0/1  
no sh  
ip add 43.43.3.1 255.255.255.0
```

R1

Routing

```
router os 1  
router-id 1.1.1.1  
net 43.43.6.1 0.0.0.0 a 0  
net 43.43.51.1 0.0.0.0 a 0  
default-inf ori always
```

02 장비 설정

Router 설정

R2

```
int lo0
ip add 192.168.2.2 255.255.255.255

int lo1
ip add 192.168.22.22 255.255.255.255

int f0/0
no sh
ip add 43.43.4.2 255.255.255.0
```

R2

Routing

```
ip route 0.0.0.0 0.0.0.0 43.43.4.12
```

02 장비 설정

Router 설정

R3

```
int lo0
ip add 192.168.3.3 255.255.255.255

int lo1
ip add 192.168.33.33 255.255.255.255

int f0/0
no sh
ip add 43.43.6.3 255.255.255.0

int f0/1
no sh
ip add 43.43.10.3 255.255.255.0
```

R3

Routing

```
router os 1
router-id 3.3.3.3
net 43.43.6.3 0.0.0.0 a 0
net 43.43.10.3 0.0.0.0 a 0
```


02 장비 설정

Router 설정

R4

```
int lo0  
ip add 192.168.4.4 255.255.255.255
```

```
int f0/1  
no sh  
ip add 43.43.9.4 255.255.255.0
```

```
int f0/0  
no sh  
ip add 43.43.6.4 255.255.255.0
```

R4

Routing

```
router os 1  
router-id 4.4.4.4  
net 43.43.9.4 0.0.0.0 a 0  
net 43.43.6.4 0.0.0.0 a 0
```

02 장비 설정

Router 설정

R5

```
int lo0  
ip add 192.168.5.5 255.255.255.255
```

```
int lo2  
ip add 43.43.52.5 255.255.255.255
```

```
int f0/0  
no sh  
ip add 43.43.8.5 255.255.255.0
```

```
int f0/1  
no sh  
ip add 43.43.7.5 255.255.255.0
```

R5

Routing

```
router os 1  
router-id 5.5.5.5  
net 43.43.7.5 0.0.0.0 a 0  
net 43.43.8.5 0.0.0.0 a 0  
net 43.43.52.5 0.0.0.0 a 0
```

02 장비 설정

Router 설정

R6

```
int lo0
ip add 192.168.6.6 255.255.255.255

int f0/1
no sh
ip add 150.2.43.254 255.255.255.0
```

R6

Routing

```
route b 254
no auto
no sync
nei 150.2.43.1 remote 43
```

02 장비 설정

Switch 설정

SW1

```
int lo150
```

```
ip add 150.1.43.1 255.255.255.0
```

```
int f1/7
```

```
no sw
```

```
ip add 43.43.2.1 255.255.255.0
```

```
ip route 43.43.0.0 255.255.0.0 43.43.2.10
```

02 장비 설정

Switch 설정

SW2

```
int f1/4
no sw
ip add 43.43.9.1 255.255.255.0

int f1/6
no sw
ip add 150.2.43.1 255.255.255.0
```

SW2

Routing

```
router os 1
net 43.43.9.1 0.0.0.0 ar 0
redi ei 200 sub

router ei 200
no au
net 150.2.43.1 0.0.0.0
redi os 1 met 1 1 1 1 1
```


02 장비 설정

Switch 설정

SW3

```
int f1/3
no sw
ip add 43.43.10.1 255.255.255.0
```

```
int f1/15
no sw
ip add 150.3.43.1 255.255.255.0
```

SW3

Routing

```
router os 1
net 43.43.10.1 0.0.0.0 ar 0
redi ei 100 sub
```

```
router ei 100
no au
net 150.3.43.1 0.0.0.0
redi os 1 met 1 1 1 1 1
```

02 장비 설정

Switch 설정

SW4

```
int f1/15  
no sw  
ip add 150.3.43.254 255.255.255.0
```

SW4

Routing

```
router ei 100  
no au  
net 150.3.43.254 0.0.0.0
```

02 장비 설정

Firewall 설정

ASA1

Context C1

int g0

no sh

int g1

no sh

int g2

no sh

admin-context admin

context admin

config-u admin.cfg

ASA1

Context 생성

context c1

config-u c1.cfg

allocate-int g0

allocate-int g1

context c2

config-u c2.cfg

allocate-int g0

allocate-int g2

mac-address auto

02 장비 설정

Firewall 설정

ASA1

Context C1

```
int g1
nameif inside
ip add 43.43.2.10 255.255.255.0
```

```
int g0
nameif outside
ip add 43.43.3.10 255.255.255.0
```

```
route outside 0 0 43.43.3.1
route inside 150.1.0.0 255.255.0.0 43.43.2.1
```

```
access-l acl_o1 per icmp a a
access-g acl_o1 in int outside
```

ASA1

Context C2

```
int g2
nameif inside
ip add 43.43.4.12 255.255.255.0
```

```
int g0
nameif outside
ip add 43.43.3.12 255.255.255.0
```

```
route outside 0 0 43.43.3.1
route outside 150.1.0.0 255.255.0.0 43.43.3.10
route inside 192.168.2.0 255.255.255.0 43.43.4.2
```

```
access-l acl_o1 per icmp a a
access-g acl_o1 in int outside
```

Firewall 설정

ASA2

```
int g0
no sh

int g1
no sh

int g2
no sh

access-l acl_o1 per icmp a a
access-g acl_o1 in int outside

router os 1
net 43.43.6.0 255.255.255.0 a 0
net 43.43.7.0 255.255.255.0 a 0
```

ASA2

Redundant 기술

```
int re 1
member-int g0
member-int g2
nameif outside
ip add 43.43.6.10 255.255.255.0

int g1
nameif inside
ip add 43.43.7.10 255.255.255.0

redundant-int re1 active-mem g0
```

Firewall (NAT) 설정

정적 NAT

```
object network L2_Server  
host 43.43.52.5  
nat (inside,outside) static 43.43.6.52
```

정적 PAT

```
object network F1_Server  
host 43.43.8.5  
nat (inside,outside) static 43.43.6.8 service tcp http 8080
```

동적 PAT

```
object network Inside_PAT  
subnet 43.43.7.0 255.255.255.0  
nat (inside,outside) dynamic interface
```

Routing Table 확인

① R1# show ip route

```
R1#show ip route
```

```
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2
```

```
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
```

```
ia - IS-IS inter area, * - candidate default, U - per-user static route
```

```
o - ODR, P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
43.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
```

```
C 43.43.3.0/24 is directly connected, FastEthernet0/1
```

```
C 43.43.6.0/24 is directly connected, FastEthernet0/0
```

```
O 43.43.7.0/24 [110/20] via 43.43.6.10, 00:04:01, FastEthernet0/0
```

```
O 43.43.8.0/24 [110/30] via 43.43.6.10, 00:04:01, FastEthernet0/0
```

```
O 43.43.9.0/24 [110/20] via 43.43.6.4, 00:05:43, FastEthernet0/0
```

```
O 43.43.10.0/24 [110/20] via 43.43.6.3, 00:05:53, FastEthernet0/0
```

```
O 43.43.52.5/32 [110/21] via 43.43.6.10, 00:04:03, FastEthernet0/0
```

```
C 43.43.51.1/32 is directly connected, Loopback2
```

```
192.168.1.0/32 is subnetted, 1 subnets
```

```
C 192.168.1.1 is directly connected, Loopback0
```

```
150.2.0.0/24 is subnetted, 1 subnets
```

```
O E2 150.2.43.0 [110/20] via 43.43.6.4, 00:05:45, FastEthernet0/0
```

```
150.3.0.0/24 is subnetted, 1 subnets
```

```
O E2 150.3.43.0 [110/20] via 43.43.6.3, 00:05:57, FastEthernet0/0
```

Routing Table 확인

② FW1(config)# show context

```
FW1(config)# show context
Context Name      Class      Interfaces      URL
*admin            default
c1                default    GigabitEthernet0,
                  GigabitEthernet1
c2                default    GigabitEthernet0,
                  GigabitEthernet2
```

Total active Security Contexts: 3

③ FW1(config)# show run context

```
FW1# sh run context

admin-context admin
context admin
  config-url disk0:/admin.cfg
!

context c1
  allocate-interface GigabitEthernet0
  allocate-interface GigabitEthernet1
  config-url disk0:/c1.cfg
!

context c2
  allocate-interface GigabitEthernet0
  allocate-interface GigabitEthernet2
  config-url disk0:/c2.cfg
!
```


Routing Table 확인

④ FW1/C1(config)# show route

```
FW1/c1(config)# show route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is 43.43.3.1 to network 0.0.0.0
```

```
C    43.43.2.0 255.255.255.0 is directly connected, inside
C    43.43.3.0 255.255.255.0 is directly connected, outside
S    150.1.0.0 255.255.0.0 [1/0] via 43.43.2.1, inside
S*   0.0.0.0 0.0.0.0 [1/0] via 43.43.3.1, outside
```

FW1/C2(config)# show route

```
FW1/c2(config)# show route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is 43.43.3.1 to network 0.0.0.0
```

```
C    43.43.3.0 255.255.255.0 is directly connected, outside
C    43.43.4.0 255.255.255.0 is directly connected, inside
S    192.168.2.0 255.255.255.0 [1/0] via 43.43.4.2, inside
S    150.1.0.0 255.255.0.0 [1/0] via 43.43.3.10, outside
S*   0.0.0.0 0.0.0.0 [1/0] via 43.43.3.1, outside
```

Routing Table 확인

⑤ FW1(config)# show mode

```
FW1(config)# show mode
Security context mode: multiple
```

⑥ FW1(config)# show run access-list c1

```
FW1/c1(config)# show run access-list
access-list acl_o1 extended permit icmp any any
```

FW1(config)# show run access-list c2

```
FW1/c2(config)# show run access-list
access-list acl_o1 extended permit icmp any any
```

Routing Table 확인

⑦ FW2(config)# show int redunant 1

```
FW2(config)# show int redundant 1
Interface Redundant1 "outside", is up, line protocol is up
  Hardware is Linux Ethernet Dev, BW 100 Mbps, DLY 100 usec
    (Full-duplex), (100 Mbps)
    Input flow control is unsupported, output flow control is unsupported
    MAC address 0000.ab06.c300, MTU 1500
    IP address 43.43.6.10, subnet mask 255.255.255.0
    904 packets input, 115268 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 pause input, 0 resume input
    511 L2 decode drops
    200 packets output, 19060 bytes, 0 underruns
    0 pause output, 0 resume output
    0 output errors, 0 collisions, 0 interface resets
    0 late collisions, 0 deferred
    0 input reset drops, 0 output reset drops
    input queue (blocks free curr/low): hardware (0/0)
    output queue (blocks free curr/low): hardware (0/0)
Traffic Statistics for "outside":
  393 packets input, 47302 bytes
  200 packets output, 16260 bytes
  52 packets dropped
  1 minute input rate 0 pkts/sec, 43 bytes/sec
  1 minute output rate 0 pkts/sec, 7 bytes/sec
  1 minute drop rate, 0 pkts/sec
  5 minute input rate 0 pkts/sec, 43 bytes/sec
  5 minute output rate 0 pkts/sec, 7 bytes/sec
  5 minute drop rate, 0 pkts/sec
Redundancy Information:
  Member GigabitEthernet0(Active), GigabitEthernet2
  Last switchover at 00:36:17 UTC Dec 19 2024
```


Routing Table 확인

⑧ FW2(config)# show route

```
FW2(config)# show route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is 43.43.6.1 to network 0.0.0.0
```

```
C    43.43.6.0 255.255.255.0 is directly connected, outside
C    43.43.7.0 255.255.255.0 is directly connected, inside
O    43.43.8.0 255.255.255.0 [110/20] via 43.43.7.5, 0:13:21, inside
O    43.43.9.0 255.255.255.0 [110/20] via 43.43.6.4, 0:13:21, outside
O    43.43.10.0 255.255.255.0 [110/20] via 43.43.6.3, 0:13:21, outside
O    43.43.52.5 255.255.255.255 [110/11] via 43.43.7.5, 0:13:21, inside
O    43.43.51.1 255.255.255.255 [110/11] via 43.43.6.1, 0:13:21, outside
O E2 150.2.43.0 255.255.255.0 [110/20] via 43.43.6.4, 0:13:21, outside
O E2 150.3.43.0 255.255.255.0 [110/20] via 43.43.6.3, 0:13:21, outside
O*E2 0.0.0.0 0.0.0.0 [110/1] via 43.43.6.1, 0:13:21, outside
```

Routing Table 확인

⑨ FW2(config)# show mode

```
FW2(config)# show mode
Security context mode: single
```

⑩ FW2(config)# show nat

```
FW2(config)# show nat

Auto NAT Policies (Section 2)
1 (inside) to (outside) source static F1_Server 43.43.6.8  service tcp www 8080
   translate_hits = 0, untranslate_hits = 0
2 (inside) to (outside) source static L2_Server 43.43.6.52
   translate_hits = 0, untranslate_hits = 0
3 (inside) to (outside) source dynamic Inside_PAT interface
   translate_hits = 0, untranslate_hits = 0
```

⑪ FW2(config)# show xlate

```
FW2(config)# show xlate
2 in use, 2 most used
Flags: D - DNS, i - dynamic, r - portmap, s - static, I - identity, T - twice
TCP PAT from inside:43.43.8.5 80-80 to outside:43.43.6.8 8080-8080
      flags sr idle 0:15:49 timeout 0:00:00
NAT from inside:43.43.52.5 to outside:43.43.6.52
      flags s idle 0:27:41 timeout 0:00:00
```

감사합니다

Thank You

안웅렬

권택

박소정

윤승원

황준서